Measure Theoretic Probability
Lesson 3
Measures, Measure Spaces and Probability Spaces

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Outline

1. Preliminaries on Sigma-Fields
2. Measures and Measure Spaces
3. Probability Measures and Probability Spaces
4. Fundamental Properties of Measures
5. Restricted Sigma-Fields and Borel Sets of the Unit Interval
6. Examples of Probability Spaces
7. Exercises and Outlook

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Objective and Setting

Setting

Let Ω be a non-empty set. We wish to assign a "size" to subsets of Ω. When Ω is uncountable or high-dimensional, assigning a size to every subset is impractical.

Strategy

1. Restrict attention to a distinguished class of subsets: a σ-field $\mathcal{F}$.
2. Define the measure on the sets of $\mathcal{F}$.
3. Specify the measure on simple sets and extend it in a natural way.

Goals of this lesson

Define a **measure**, a **measure space** and a **probability space**; derive their basic properties; construct two examples; and describe the Borel σ-field on $[0, 1]$.

Recall: σ-Fields and Generated σ-Fields

Definition: σ-field

A collection $\mathcal{F}$ of subsets of Ω is a **σ-field** on Ω if

(i) $\Omega \in \mathcal{F}$;

(ii) $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$;

(iii) $A_1, A_2, \cdots \in \mathcal{F} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

Definition: generated σ-field

For a collection of subsets of Ω, denotes the **smallest σ-field containing A** .

Why these collections?

Measures will be defined on σ-fields. We now need two further facts about how σ-fields combine **Unions** and **Intersections**.

Unions of σ-Fields

Proposition

If $\mathcal{F}_1$ and $\mathcal{F}_2$ are σ-fields on Ω, then $\mathcal{F}_1 \cup \mathcal{F}_2$ need **not** be a σ-field.

Counterexample

Let $A, B \subseteq \Omega$ and

$$\mathcal{F}_1 = A = \{\emptyset, A, A^c, \Omega\}, \qquad \mathcal{F}_2 = B = \{\emptyset, B, B^c, \Omega\}.$$

Then $\mathcal{F}_1 \cup \mathcal{F}_2 = \{\emptyset, A, A^c, B, B^c, \Omega\}$. For $\Omega = \{1, 2, 3\}$, $A = \{1\}$, $B = \{2\}$ we get $A \cup B = \{1, 2\} \notin \mathcal{F}_1 \cup \mathcal{F}_2$, so (iii) fails.

Remark

If $A \subseteq B$ then $A \cup B = B$ is present, but other required sets can still be missing. For $A = \{1\}$, $B = \{1, 2\}$, the set $A \cup B^c = \{1, 3\}$ is absent.

Intersections of Arbitrary Families of σ-Fields

Proposition

Let I be **any** index set (finite, countably infinite or uncountable), and let $\mathcal{F}_i$ be a σ-field on Ω for each $i \in I$. Then

$$\bigcap_{i \in I} \mathcal{F}_i$$

is a σ-field on Ω.

Why a direct proof is needed

Induction from the case of two σ-fields gives only *finite* intersections. Passing from "finite" to "countable" is not automatic: the field on $\mathbb{R}$ in Lesson 1 was closed under finite unions, yet a countable union of its sets left it.

Proof: Intersections Preserve σ-Fields

Write $\mathcal{G} = \bigcap_{i \in I} \mathcal{F}_i$.

Proof

(i) $\Omega \in \mathcal{F}_i$ for every i, hence $\Omega \in \mathcal{G}$.

(ii) Let $A \in \mathcal{G}$. Then $A \in \mathcal{F}_i$ for every i, so $A^c \in \mathcal{F}_i$ for every i, hence $A^c \in \mathcal{G}$.

(iii) Let $A_1, A_2, \cdots \in \mathcal{G}$. For each fixed i, all A_n lie in $\mathcal{F}_i$, and $\mathcal{F}_i$ is a σ-field, so $\bigcup_n A_n \in \mathcal{F}_i$. This holds for every i, hence $\bigcup_n A_n \in \mathcal{G}$. □

Caution

A set belongs to $\bigcap_i \mathcal{F}_i$ when the **whole set** is a member of every collection $\mathcal{F}_i$. We never intersect individual members of different σ-fields.

Consequence: The Generated σ-Field Is Well Defined

Formula

For a collection of subsets of Ω,

$$= \bigcap \{\mathcal{F} : \ \mathcal{F} \text{ is a } \sigma\text{-field on } \Omega, \ \subseteq \mathcal{F}\}.$$

Justification

- The family is non-empty: it contains the power set $\mathcal{P}(\Omega)$.
- The intersection is a σ-field, by the preceding proposition (the family may be uncountable).
- It contains , and is contained in every σ-field containing : it is the **smallest** one.

Measurable Sets and Measurable Spaces

Definition: measurable set

If $\mathcal{F}$ is a σ-field on Ω, the members of $\mathcal{F}$ are called **measurable sets**.

Definition: measurable space

A pair $(\Omega, \mathcal{F})$, where Ω is a non-empty set and $\mathcal{F}$ is a σ-field on Ω, is called a **measurable space**.

Convention

From now on, statements begin with "Let $(\Omega, \mathcal{F})$ be a measurable space", which stands for: Ω is a non-empty set and $\mathcal{F}$ is a σ-field on Ω.

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Definition of a Measure

Definition

Let $(\Omega, \mathcal{F})$ be a measurable space. A **measure** on $(\Omega, \mathcal{F})$ is a function

$$\mu : \mathcal{F} \to [0, \infty]$$

such that

(M1) $\mu(\emptyset) = 0$;

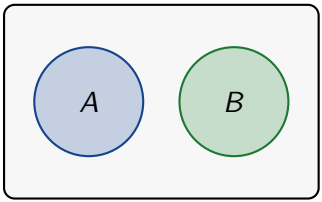
(M2) **countable additivity**: for pairwise disjoint $A_1, A_2, \cdots \in \mathcal{F}$,

$$\mu\Big(\bigcup_{n=1}^{\infty} A_n\Big) = \sum_{n=1}^{\infty} \mu(A_n).$$

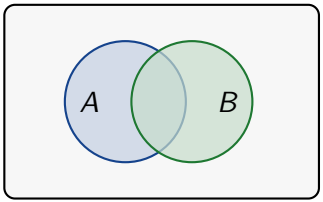
Remarks

- $[0, \infty] = [0, \infty) \cup \{\infty\}$ (non-negative extended reals): a measure may be 0, positive or infinite.

Interpretation of Additivity (Area)



Disjoint sets: the area of the union is the sum of the areas.



Overlapping sets: adding the areas counts $A \cap B$ twice.

Reading the definition

A measure takes a set of $\mathcal{F}$ and returns a number in $[0, \infty]$. Condition (M2) formalises the left-hand picture; for overlapping sets, only an inequality can be expected.

Finite Additivity

Proposition

Let μ be a measure on $(\Omega, \mathcal{F})$ and $A_1, \ldots, A_n \in \mathcal{F}$ pairwise disjoint. Then

$$\mu\Big(\bigcup_{k=1}^{n} A_k\Big) = \sum_{k=1}^{n} \mu(A_k).$$

Proof

Put $A_k = \emptyset$ for all $k > n$. The empty set is disjoint from every set, including itself ($\emptyset \cap A = \emptyset$), so $A_1, A_2, \ldots$ are pairwise disjoint, and the union is unchanged. By (M2) and (M1),

$$\mu\Big(\bigcup_{k=1}^{n} A_k\Big) = \sum_{k=1}^{n} \mu(A_k) + \sum_{k>n} \mu(\emptyset) = \sum_{k=1}^{n} \mu(A_k). \qquad \square$$

An Equivalent Axiom in Place of $\mu(\emptyset) = 0$

Proposition

Let $\mu : \mathcal{F} \to [0, \infty]$ satisfy (M2) and **(M1′):** there exists $A \in \mathcal{F}$ with $\mu(A) < \infty$. Then $\mu(\emptyset) = 0$, so μ is a measure. (Conversely, (M1) gives (M1′) since $\mu(\emptyset) = 0 < \infty$.)

Proof

The sequence $A, \emptyset, \emptyset, \ldots$ is pairwise disjoint with union A. By (M2),

$$\mu(A) = \mu(A) + \sum_{k=2}^{\infty} \mu(\emptyset).$$

Since $\mu(A) < \infty$ we may subtract it: $\sum_{k \geq 2} \mu(\emptyset) = 0$. A series of non-negative terms with sum 0 has all terms 0, so $\mu(\emptyset) = 0$. □

Why finiteness is needed

If $\mu(A) = \infty$, the equation $\infty = \infty + c$ holds for every c, so nothing can be concluded.

Notation and Standard Examples of Measures

Notation

Symbol	Meaning
$\mu(\cdot)$	a generic measure
$\mathbb{P}(\cdot)$	a probability measure
$\lambda(\cdot)$	Lebesgue measure

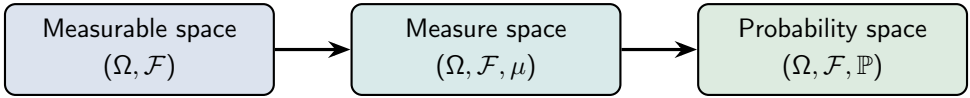
Lebesgue measure on $\mathbb{R}$

On $(\mathbb{R}, (\mathbb{R}))$, where $(\mathbb{R})$ is the Borel σ-field, the Lebesgue measure of an interval is its length:

$$\lambda\big((a,b)\big) = b - a \qquad (a < b).$$

In an n-dimension, it is the n-dimensional volume (length for $n = 1$, area for $n = 2$, volume for $n = 3$)., $\ldots$ (R^n)

Measurable Spaces, Measure Spaces, Probability Spaces



Definitions

- A **measure space** is a triple $(\Omega, \mathcal{F}, \mu)$: a measurable space together with a measure μ on $\mathcal{F}$.
- A **probability space** is a measure space whose measure is a **probability measure**.

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Definition of a Probability Measure

Definition

Let $(\Omega, \mathcal{F})$ be a measurable space. A **probability measure** is a measure

$$\mathbb{P} : \mathcal{F} \to [0,1] \qquad \text{with} \qquad \mathbb{P}(\Omega) = 1.$$

Explicitly: $\mathbb{P}(\emptyset) = 0$, $\mathbb{P}$ is countably additive on pairwise disjoint sets of $\mathcal{F}$, and $\mathbb{P}(\Omega) = 1$. The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is a **probability space**.

Dictionary with elementary probability

Elementary	Measure-theoretic
sample space	the set Ω
event	a measurable set $A \in \mathcal{F}$
probability of an event	$\mathbb{P}(A)$

Complement Rule

Proposition

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $A \in \mathcal{F}$. Then

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

Proof

The sets A and A^c belong to $\mathcal{F}$, are disjoint, and $A \cup A^c = \Omega$. By finite additivity,

$$1 = \mathbb{P}(\Omega) = \mathbb{P}(A \cup A^c) = \mathbb{P}(A) + \mathbb{P}(A^c),$$

which gives the claim. □

Remark

The familiar rule from elementary probability is a **consequence** of the definition; it need not be assumed.

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Monotonicity

Proposition

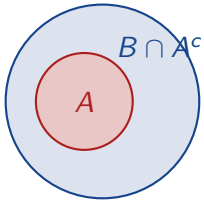
Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and $A, B \in \mathcal{F}$ with $A \subseteq B$. Then $\mu(A) \leq \mu(B)$.

Proof

Write $B = (B \cap A) \cup (B \cap A^c) = A \cup (B \cap A^c)$, a union of two disjoint sets of $\mathcal{F}$. By finite additivity,

$$\mu(B) = \mu(A) + \mu(B \cap A^c) \;\geq\; \mu(A),$$

since $\mu(B \cap A^c) \geq 0$. □



Strict inclusion does not give strict inequality

For $A \subsetneq B$ one may have $\mu(A) = \mu(B)$. Example: $\lambda((0,1]) = \lambda([0,1]) = 1$.

Set Difference

Definition

For $A, B \in \mathcal{F}$,

$$A \setminus B := A \cap B^c.$$

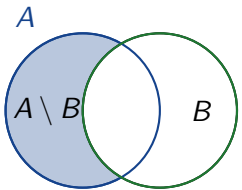
This set lies in $\mathcal{F}$ ($\mathcal{F}$ is closed under complements and intersections).

Facts

(a) $A \setminus B \subseteq A$, so $\mu(A \setminus B) \leq \mu(A)$ by monotonicity.

(b) By De Morgan's law,

$$A \setminus (B_1 \cup B_2) = A \cap B_1^c \cap B_2^c.$$



Countable Subadditivity

Proposition

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and $A_1, A_2, \cdots \in \mathcal{F}$, **not necessarily disjoint**. Then

$$\mu\Big(\bigcup_{n=1}^{\infty} A_n\Big) \leq \sum_{n=1}^{\infty} \mu(A_n).$$

Idea

For disjoint sets (M2) gives equality. When sets overlap, summing the individual measures counts the overlaps more than once (see the area picture), so the sum can only be larger.

Proof strategy

Rewrite $\bigcup_n A_n$ as a **disjoint** union of measurable pieces $B_n \subseteq A_n$, apply (M2) to the pieces, and compare each $\mu(B_n)$ with $\mu(A_n)$ by monotonicity.

Proof of Countable Subadditivity: Disjointification

Construction

$B_1 = A_1$ and, for $n \geq 2$,

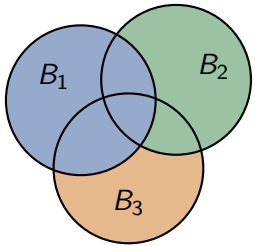
$$B_n = A_n \setminus (A_1 \cup \cdots \cup A_{n-1}) = A_n \cap A_1^c \cap \cdots \cap A_{n-1}^c.$$

Properties

(a) $B_n \in \mathcal{F}$ and $B_n \subseteq A_n$.

(b) The B_n are pairwise disjoint: if $m < n$, then $B_m \subseteq A_m$ and $B_n \subseteq A_m^c$.

(c) $\bigcup_n B_n = \bigcup_n A_n$: if $x \in \bigcup_n A_n$ and n is the smallest index with $x \in A_n$, then $x \in B_n$.



A_1. $A_2 \setminus A_1$. Orange: $A_3 \setminus (A_1 \cup A_2)$.

Proof of Countable Subadditivity: Conclusion

Proof (continued)

By properties (c), (b) and (M2), then (a) with monotonicity,

$$\mu\Big(\bigcup_{n=1}^{\infty} A_n\Big) = \mu\Big(\bigcup_{n=1}^{\infty} B_n\Big) = \sum_{n=1}^{\infty} \mu(B_n) \leq \sum_{n=1}^{\infty} \mu(A_n).$$

Remark

Taking $A_n = \emptyset$ for $n > 2$ gives $\mu(A \cup B) \leq \mu(A) + \mu(B)$, the finite version.

Summary of Properties of a Measure

Consequences of the definition

Property	Statement	Ingredients
Finite additivity	disjoint $A_1, \ldots, A_n$: $\mu(\bigcup A_k) = \sum \mu(A_k)$	(M1), (M2), padding by $\emptyset$
Complement rule	$\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$	finite additivity, $\mathbb{P}(\Omega) = 1$
Monotonicity	$A \subseteq B \Rightarrow \mu(A) \leq \mu(B)$	finite additivity, $\mu \geq 0$
Countable subadditivity	$\mu(\bigcup A_n) \leq \sum \mu(A_n)$	disjointification, (M2), monotonicity

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Motivation: A Probability Space on $[0, 1]$

Goal

Use Lebesgue measure as a probability measure on subsets of $[0, 1]$. For this we need a σ-field on $\Omega = [0, 1]$.

The difficulty

The natural candidate is the σ-field generated by the open sets *of* $[0, 1]$. Openness in a subspace behaves differently from openness in $\mathbb{R}$: for instance $[0, 1)$ is neither open nor closed in $\mathbb{R}$, but it is open in $[0, 1]$. Instead of developing the topology, we obtain this σ-field by **restricting** the Borel sets of $\mathbb{R}$ to $[0, 1]$.

Plan

1. Show that restricting any σ-field to a subset Ω' gives a σ-field on Ω'.
2. Show that restriction commutes with taking generated σ-fields.
3. Conclude: both constructions of the Borel sets of $[0, 1]$ agree.

Restriction of a σ-Field

Proposition

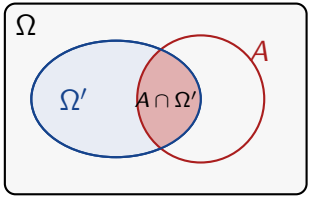
Let $\mathcal{F}$ be a σ-field on Ω and $\Omega' \subseteq \Omega$ non-empty. Then

$$\mathcal{F}' := \{A \cap \Omega' : \ A \in \mathcal{F}\}$$

is a σ-field **on** Ω'.

Note

Complements are now taken **with respect to** Ω': for $C \subseteq \Omega'$, $C^c = \Omega' \setminus C$.



Proof: Restriction Gives a σ-Field on Ω'

Proof

(i) $\Omega \in \mathcal{F}$, so $\Omega' = \Omega \cap \Omega' \in \mathcal{F}'$.

(ii) Let $A' = A \cap \Omega' \in \mathcal{F}'$ with $A \in \mathcal{F}$. The complement of A' in Ω' is

$$\Omega' \setminus (A \cap \Omega') = A^c \cap \Omega',$$

and $A^c \in \mathcal{F}$, so it belongs to $\mathcal{F}'$.

(iii) Let $A'_n = A_n \cap \Omega' \in \mathcal{F}'$ with $A_n \in \mathcal{F}$. Then

$$\bigcup_n A'_n = \left(\bigcup_n A_n\right) \cap \Omega',$$

and $\bigcup_n A_n \in \mathcal{F}$, so the union belongs to $\mathcal{F}'$. □

Restriction Commutes with Generation

Proposition

Let be a collection of subsets of Ω (not necessarily a σ-field) and $\Omega' \subseteq \Omega$ non-empty. Put $A' = \{A \cap \Omega' : A \in\}$. Then

$$\sigma_{\Omega'}(') = \{A \cap \Omega' : \ A \in \},$$

where $\sigma_{\Omega'}(')$ is the σ-field on Ω' generated by A'.

In words

Restricting the sets first and then generating a σ-field gives the same result as generating first and then restricting.

Proof: Restriction Commutes with Generation

($\subseteq$) $\mathcal{F}' \subseteq \sigma_{\Omega'}(')$

Define $\mathcal{G} = \{A \subseteq \Omega : \ A \cap \Omega' \in \sigma_{\Omega'}(')\}$. This is a σ-field on Ω: it contains Ω (as $\Omega \cap \Omega' = \Omega'$); if $A \in \mathcal{G}$ then $A^c \cap \Omega' = \Omega' \setminus (A \cap \Omega') \in \sigma_{\Omega'}(')$; and $(\bigcup A_n) \cap \Omega' = \bigcup (A_n \cap \Omega')$ handles unions. Also $\subseteq \mathcal{G}$. So $= \mathcal{F} \subseteq \mathcal{G}$, i.e. $A \cap \Omega' \in \sigma_{\Omega'}(')$ for all $A \in \mathcal{F}$. $\square$

Borel Sets of $[0,1]$

Definition

The **Borel σ-field on** $[0,1]$ is the restriction of $(\mathbb{R})$ to $[0,1]$:

$$([0,1]) = \{B \cap [0,1] : \ B \in (\mathbb{R})\}.$$

Theorem (two descriptions agree)

$([0,1])$ is also the σ-field on $[0,1]$ generated by the **open subsets of** $[0,1]$, i.e. by the sets $U \cap [0,1]$ with U open in $\mathbb{R}$.

Proof

Take $\Omega = \mathbb{R}$, $\Omega' = [0,1]$, and $\ $ the collection of all open subsets of $\mathbb{R}$. Since every open set is a countable union of open intervals, $\ = (\mathbb{R})$. Then $'$ is the collection of open subsets of $[0,1]$, and the preceding proposition gives $\sigma_{[0,1]}(') = ([0,1])$. □

Example

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Example 1: Three Coin Tosses

Setup

$\Omega =$
$\{HHH, HHT, HTH, HTT, THH, THT, T$
(8 outcomes), $\mathcal{F} = \mathcal{P}(\Omega)$.
For a fair coin, $\mathbb{P}(\{\omega\}) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$
for each outcome ω.

Extending from single outcomes to sets

For example, with $A = \{HHH, HHT\}$:

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(\{HHH\} \cup \{HHT\}) \\ &= \mathbb{P}(\{HHH\}) + \mathbb{P}(\{HHT\}) \quad \text{(finite additivity)} \\ &= \tfrac{1}{8} + \tfrac{1}{8} = \tfrac{1}{4}.\end{aligned}$$

In general, every $A \in \mathcal{F}$ is a finite disjoint union of singletons, so $\mathbb{P}(A) = |A|/8$.

Principle

Probabilities are assigned to the simplest sets (single outcomes) and **extended** to all of $\mathcal{F}$ by additivity: exactly the approach announced at the start of the lesson.

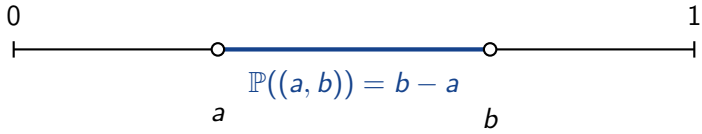
Example 2: Lebesgue Measure on $[0, 1]$

Setup

$\Omega = [0, 1]$, $\mathcal{F} = ([0, 1])$, and $\mathbb{P} = \lambda$ restricted to $\mathcal{F}$.

Verification that $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space

- λ is a measure, so $\mathbb{P}$ satisfies (M1) and (M2).
- $\mathbb{P}(\Omega) = \lambda([0, 1]) = 1 - 0 = 1$.
- For every $A \in \mathcal{F}$, $A \subseteq [0, 1]$, so by monotonicity $0 \leq \mathbb{P}(A) \leq \mathbb{P}(\Omega) = 1$: the values lie in $[0, 1]$.



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Exercises

1. Let $\Omega = \{1, 2, 3\}$, $\mathcal{F}_1 = \{1\}$, $\mathcal{F}_2 = \{2\}$. Determine $\mathcal{F}_1 \cup \mathcal{F}_2$.
2. Let $A, B \in \mathcal{F}$ with $\mu(A \cap B) < \infty$. Prove $\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B)$.
3. For a probability measure and $A \subseteq B$ in $\mathcal{F}$, prove $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$.
4. Show that $\{0\}$ and $(0, \frac{1}{2}]$ belong to $([0, 1])$.

Summary

Key definitions

Object	Definition
Measurable space	$(\Omega, \mathcal{F})$ with $\mathcal{F}$ a σ-field on Ω
Measure	$\mu : \mathcal{F} \to [0, \infty]$, $\mu(\emptyset) = 0$, countably additive on disjoint sets
Probability measure	a measure $\mathbb{P} : \mathcal{F} \to [0, 1]$ with $\mathbb{P}(\Omega) = 1$
Measure / probability space	$(\Omega, \mathcal{F}, \mu)$ / $(\Omega, \mathcal{F}, \mathbb{P})$
Borel sets of $[0, 1]$	$([0, 1]) = \{B \cap [0, 1] : B \in (\mathbb{R})\}$

Key results

Finite additivity, complement rule, monotonicity, countable subadditivity; intersections of σ-fields are σ-fields, but the same argument does not necessarily hold for unions in general.

Next Lesson

Extension of a probability measure

Specifying a measure on every set of a σ-field is impractical. We will start with a probability measure defined only on a **field** $\mathcal{F}_0$ and extend it, in a natural way, to all sets of the σ-field $\mathcal{F}_0$ that it generates.